# Metric Scaling Transformations in Synthetic Spacetimes and Analog Optical Lattices

A Theoretical Framework for Boundary Impedance Matching and Horizon Preservation ($G_c = \pm 1.0857$)

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Repository: universal-constant-universal metric scaler
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Technical Abstract
Analog spacetime experiments utilizing Bose-Einstein Condensates (BECs) in optical lattices frequently encounter acoustic horizon collapse during dynamic expansion and impedance mismatches across synthetic curvature boundaries. This paper formulates the application of the invariant metric scale factor $\mathbf{G_c = 1.0857}$ and its quadratic energy counterpart $\mathbf{G_c^2 = 1.1787}$ to phonon scalar field propagation. We demonstrate that linear coordinate adjustments preserve acoustic horizon stability while quadratic energy scaling balances chemical potential steps across optical potential interfaces.

## 1. Acoustic Spacetime Formulation

In ultracold atomic condensate experiments, density perturbations (phonons) behave as scalar fields $\varphi$ propagating through an effective d'Alembertian curved acoustic metric $g_{\mu\nu}$:

$$(1 / \sqrt{(-g)})\, \partial_\mu [\, \sqrt{(-g)}\, g^{\mu\nu}\, \partial_\nu \varphi\, ] = 0$$

The effective light speed within the acoustic spacetime is dictated by the local speed of sound $c_s = \sqrt{(g\, n_0 / m)}$. As the trap potential expands to simulate cosmological inflation, particle density $n_0$ decays, causing $c_s$ to collapse below thermal noise thresholds before horizon freeze-out phenomena can be resolved.

## 2. Comparative Scaling Parameters

Enforcing the transformation metric multiplier $G_c$ across sound speed profiles, cosmological scale factors, and chemical interaction potentials adjusts key laboratory parameters as outlined below:

| Synthetic Metric Parameter | Standard Laboratory Baseline | $G_c$ Scaled Target (1.0857) | Physical Impact / Problem Resolved |
|---|---|---|---|
| **Acoustic Light Speed ($c_s$)** | $c_{s,\,0}$ (Density dependent) | $1.0857 \times c_{s,\,0}$ (+8.57%) | Prevents signal dissipation during fast trap expansions. |
| **Scale Factor Profile ($a(t)$)** | $a_0(t)$ (Trap boundary) | $1.0857 \times a_0(t)$ (+8.57%) | Extends lookback horizon for simulated cosmic inflation. |
| **Interaction Energy ($\mu \propto c_s^2$)** | $\mu_0$ (Chemical potential) | $1.1787 \times \mu_0$ (+17.87%) | Eliminates impedance reflection at metric step jumps. |
| **Phonon Kinetic** | $E_{k,\,0}$ | $1.1787 \times E_{k,\,0}$ | Maintains stress-energy momentum |

| Synthetic Metric Parameter | Standard Laboratory Baseline | $G_c$ Scaled Target (1.0857) | Physical Impact / Problem Resolved |
|---|---|---|---|
| **Energy ($E_k$)** | | ($G_c^2$) | flux across boundaries. |



# 3. Resolution of Horizon Collapse and Reflection Artifacts

When optical lattices are dynamically deformed to model curved expanding geometries (such as de Sitter or Anti-de Sitter spaces), two primary artifacts degrade observational fidelity: thermal decoherence and boundary reflection.

## A. Acoustic Horizon Stabilization via Linear Scaling ($G_c$)

Applying a linear coordinate transformation $G_c = 1.0857$ to the trap expansion rate velocity $v_0(r, t)$ and local sound speed $c_s(r, t)$ expands the effective acoustic horizon footprint. This linear boost prevents the effective speed of light from falling below the thermal fluctuation floor, allowing vacuum phonon pair creation (analog Hawking/cosmological radiation) to freeze out cleanly across synthetic horizons.

## B. Boundary Impedance Matching via Quadratic Scaling ($G_c^2$)

Step-function changes in optical potential wells create unphysical wave reflections for scalar field excitations. Because kinetic energy and the stress-energy tensor scale quadratically with velocity ($E_k \propto v^2$), applying $G_c^2 \approx 1.1787$ to the background chemical potential $\mu$ balances the momentum flux across flat-to-curved metric transitions, ensuring continuous phase transmission without reflection spikes.